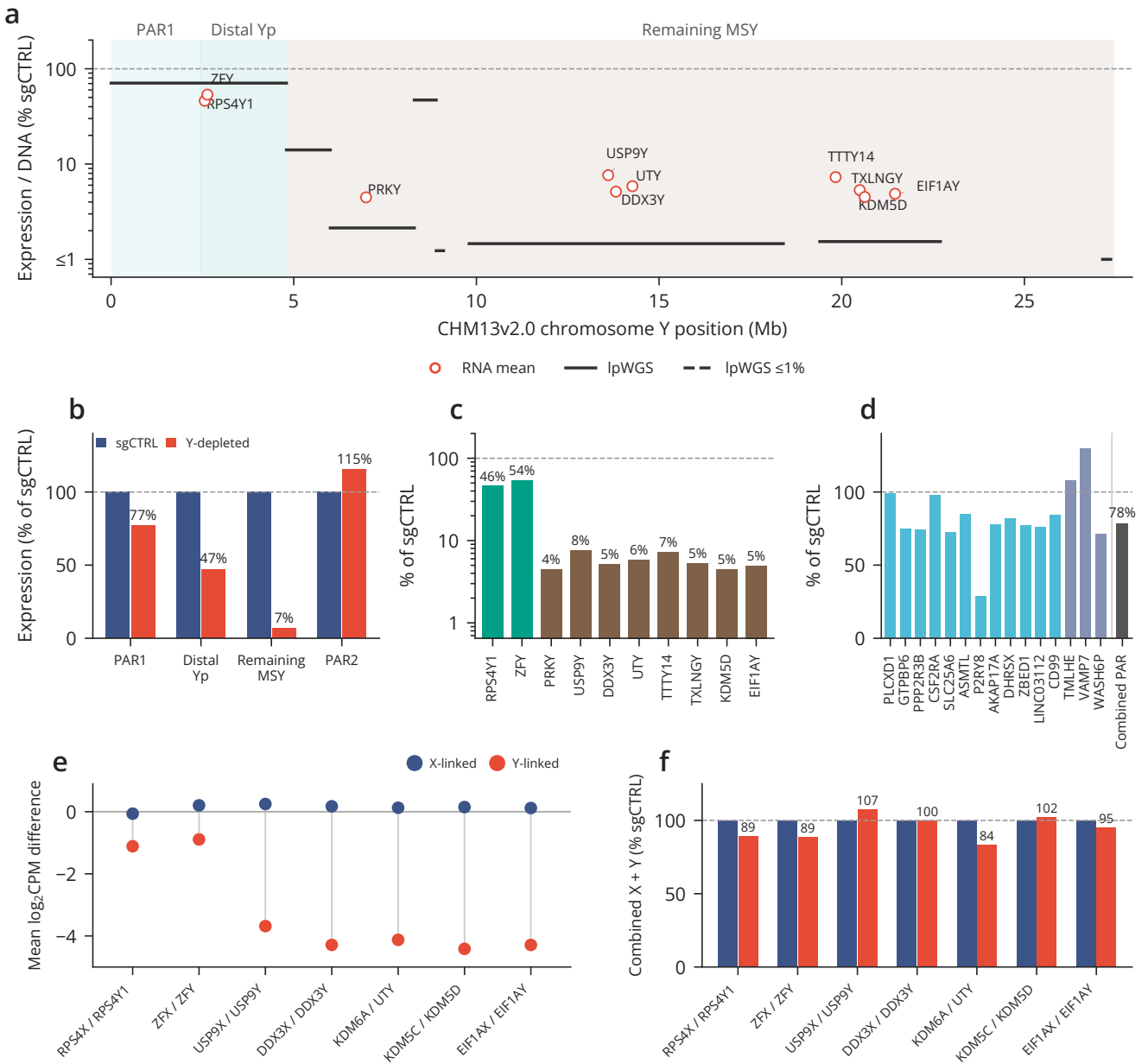
a
PAR1
Distal Yp
Remaining MSY
Expression / DNA (% sgCTRL)
100
10
≤1
ZFY
RPS4Y1
PRKY
USP9Y
UTY
DDX3Y
TTTY14
TXLNGY
EIF1AY
KDM5D
0
5
10
15
20
25
CHM13v2.0 chromosome Y position (Mb)
RNA mean
lpWGS
lpWGS ≤1%
b
sgCTRL
Y-depleted
Expression (% of sgCTRL)
100
50
0
77%
47%
7%
115%
PAR1
Distal Yp
Remaining MSY
PAR2
c
% of sgCTRL
100
10
1
46%
54%
4%
8%
5%
6%
7%
5%
5%
5%
RPS4Y1
ZFY
PRKY
USP9Y
DDX3Y
UTY
TTTY14
TXLNGY
KDM5D
EIF1AY
d
% of sgCTRL
100
50
0
78%
PLCXD1
GTPBP6
PPP2R3B
CSF2RA
SLC25A6
ASMTL
P2RY8
AKAP17A
DHRSX
ZBED1
LINC03112
CD99
TMLHE
VAMP7
WASH6P
Combined PAR
e
X-linked
Y-linked
Mean log2CPM difference
0
−2
−4
RPS4X / RPS4Y1
ZFX / ZFY
USP9X / USP9Y
DDX3X / DDX3Y
KDM6A / UTY
KDM5C / KDM5D
EIF1AX / EIF1AY
f
Combined X + Y (% sgCTRL)
100
50
0
89
89
107
100
84
102
95
RPS4X / RPS4Y1
ZFX / ZFY
USP9X / USP9Y
DDX3X / DDX3Y
KDM6A / UTY
KDM5C / KDM5D
EIF1AX / EIF1AY